

DL Midsem Notes — Section 4

Perceptron Model and Convergence Proof

1 Theory

1.1 The Perceptron Model

$$\hat{y} = \text{sign}(w \cdot x + b) = \begin{cases} 1 & w \cdot x + b \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

Unlike the MP neuron, w is real-valued and **learned**. Geometrically, $w \cdot x + b = 0$ is a hyperplane; the perceptron can only correctly classify data that some hyperplane can split cleanly — this is **linear separability**.

1.2 The Learning Rule

For a misclassified example (x, y) , $y \in \{+1, -1\}$:

$$w \leftarrow w + \eta y x, \quad b \leftarrow b + \eta y$$

Update only happens on mistakes; correctly classified points leave w unchanged. Intuitively, this nudges the decision boundary toward correctly classifying the point that was just missed.

1.3 Linear Separability

Two classes are linearly separable if a single hyperplane divides them with no misclassifications. AND, OR, NOT are linearly separable. XOR is the classic counterexample (proved in Q1) — this is exactly the limitation that killed single-layer perceptrons in the first AI winter.

1.4 Perceptron Convergence Theorem (Statement)

If a dataset is linearly separable with margin $\gamma > 0$ (i.e. a unit vector w^* exists with $y_i(w^* \cdot x_i) \geq \gamma$ for every point), and every input satisfies $\|x_i\| \leq R$, then the perceptron learning algorithm makes at most R^2/γ^2 mistakes (updates) before it finds a separating hyperplane and stops. It is guaranteed to terminate in finite steps — full proof in Q2.

2 Practice Questions

Q1 (PYQ-style, CS6510-1 Q2A). Determine whether it is possible to implement the XOR gate using a single perceptron. Prove your answer.

Q2 (PYQ-style, CS6510-1 Q1B / CS6510-2 Q6 / CS6510-9 Q3b — asked in 3 separate papers, high priority). If the sets P and N are finite and linearly separable, prove that the perceptron learning algorithm updates the weight vector w_t a finite number of times such that the two sets are separated.

Q3 (PYQ-style, CS6510-5 Q5b). What is the minimum number of perceptron units sufficient to model the 3-variable XOR (parity) function using a single hidden layer and an output layer? Sketch the construction.

Q4 (concept check). Give a one-line formal definition of linear separability, and state one Boolean function that satisfies it and one that doesn't.

3 Answers

A1. Not possible with a single perceptron. Suppose weights w_1, w_2 and bias b existed with $\hat{y} = 1 \iff w_1x_1 + w_2x_2 + b \geq 0$, matching XOR: $(0, 0) \rightarrow 0$, $(1, 1) \rightarrow 0$, $(0, 1) \rightarrow 1$, $(1, 0) \rightarrow 1$. Then:

$$\begin{aligned} (0, 0) \rightarrow 0 : b < 0 \quad (1, 1) \rightarrow 0 : w_1 + w_2 + b < 0 \\ (0, 1) \rightarrow 1 : w_2 + b \geq 0 \quad (1, 0) \rightarrow 1 : w_1 + b \geq 0 \end{aligned}$$

Adding the last two: $w_1 + w_2 + 2b \geq 0 \Rightarrow w_1 + w_2 \geq -2b$. Combined with $w_1 + w_2 < -b$ (from the second condition): $-2b \leq w_1 + w_2 < -b \Rightarrow -2b < -b \Rightarrow b > 0$, contradicting $b < 0$ from the first condition. No such w_1, w_2, b exist — XOR is not linearly separable, so no single perceptron can realize it.

A2 (Novikoff's proof). Let w^* (with $\|w^*\| = 1$) separate the data with margin $\gamma > 0$: $y_i(w^* \cdot x_i) \geq \gamma$ for every point. Let $R = \max_i \|x_i\|$. Let w_k be the weight vector after k mistake-driven updates, $w_0 = 0$. On the k -th mistake, (x, y) is misclassified, so $y(w_{k-1} \cdot x) \leq 0$, and the update is $w_k = w_{k-1} + yx$.

Lower bound on $w_k \cdot w^*$:

$$w_k \cdot w^* = w_{k-1} \cdot w^* + y(x \cdot w^*) \geq w_{k-1} \cdot w^* + \gamma$$

since the margin condition holds for *every* point regardless of the current weights. By induction, $w_k \cdot w^* \geq k\gamma$.

Upper bound on $\|w_k\|^2$:

$$\|w_k\|^2 = \|w_{k-1}\|^2 + 2y(w_{k-1} \cdot x) + \|x\|^2 \leq \|w_{k-1}\|^2 + \|x\|^2 \leq \|w_{k-1}\|^2 + R^2$$

using $y(w_{k-1} \cdot x) \leq 0$ (it was misclassified). By induction, $\|w_k\|^2 \leq kR^2$.

Combine via Cauchy–Schwarz:

$$\begin{aligned} k\gamma \leq w_k \cdot w^* &\leq \|w_k\| \|w^*\| = \|w_k\| \leq \sqrt{kR^2} = \sqrt{k} R \\ \Rightarrow \sqrt{k} \gamma &\leq R \Rightarrow k \leq \frac{R^2}{\gamma^2} \end{aligned}$$

The number of mistakes is bounded by a finite constant R^2/γ^2 that doesn't depend on the order of examples, so the algorithm must stop making mistakes — i.e. converge to a separating hyperplane — after finitely many updates. ■

A3. 3 hidden units suffice, using a counting construction. Let $s = x_1 + x_2 + x_3 \in \{0, 1, 2, 3\}$; parity (XOR3) is 1 exactly when s is odd ($s = 1$ or $s = 3$). Build three threshold hidden units:

$$h_1 = \mathbb{K}[s \geq 1], \quad h_2 = \mathbb{K}[s \geq 2], \quad h_3 = \mathbb{K}[s \geq 3]$$

(each is a perceptron with weights 1, 1, 1 on x_1, x_2, x_3 and threshold 0.5, 1.5, 2.5 respectively). Then the output unit computes $y = \text{sign}(h_1 - h_2 + h_3 - 0.5)$:

s	h_1	h_2	h_3	$h_1 - h_2 + h_3$
0	0	0	0	0
1	1	0	0	1
2	1	1	0	0
3	1	1	1	1

Output is 1 exactly at $s = 1, 3$ (odd) and 0 at $s = 0, 2$ (even) — correct parity, using only 3 hidden units. (General pattern: n -bit parity needs n hidden threshold units built this way.)

A4. Two classes are linearly separable if there exists a hyperplane $w \cdot x + b = 0$ such that every point of one class satisfies $w \cdot x + b > 0$ and every point of the other satisfies $w \cdot x + b < 0$. AND is linearly separable (e.g. $w = (1, 1), b = -1.5$ separates it); XOR is not (proved in Q1).